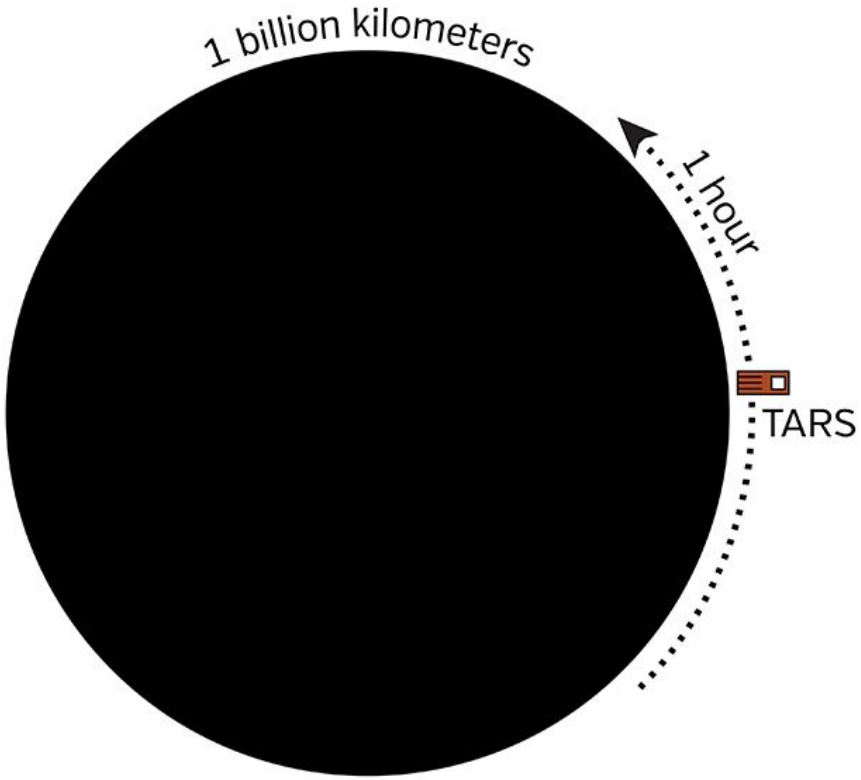


Fig. 6.2. TARS, falling into Gargantua, is dragged around the hole's billion-kilometer circumference once each hour, as seen from afar.

In 1975, I discovered a mechanism by which Nature protects black holes from spinning faster than the maximum: When it gets close to the maximum spin, a black hole has difficulty capturing objects that orbit in the same direction as the hole rotates and that therefore, when captured, increase the hole's spin. But the hole easily captures things that orbit opposite to its spin and that, when captured, slow the hole's spin. Therefore, the spin is easily slowed, when it gets close to the maximum.

In my discovery, I focused on a disk of gas, somewhat like Saturn's rings, that orbits in the same direction as the hole's spin: an *accretion disk* (Chapter 9). Friction in the disk makes the gas gradually spiral into the black hole, spinning it up. Friction also heats the gas, making it radiate photons. The whirl of space around the hole grabs those photons that travel in the same direction as the hole spins and flings them away,